

PROVISIONAL PATENT APPLICATION

Engineered Coherence Media, Phase-Coherent Cavity-Array Drive Devices, Travel-Guide Feeds, Stress-Isolating Mounts, Sealed Drive Modules, and Tiled Media Assemblies for Angular-Momentum Drive Devices, and Methods of Making, Qualifying, and Operating Them

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This specification is complete and self-contained. Applicant's companion applications are nonessential background only; no material essential to written description, enablement, or the claims is incorporated from them. Every example under PROPHETIC EXAMPLES is prospective; no experiment, nonzero force response, qualified medium class, or qualified module is represented as achieved, and no force response, sign, magnitude, additivity, or scaling law is presumed. Where this specification speaks of a force offset, it names a measured quantity that may be zero.

Portfolio prophetic-example statement: All examples identified as prophetic are prospective; no prophetic example is represented as an experiment performed or a result achieved.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is one of a coordinated series directed to applications of controlled angular-momentum coherence in condensed media (companion drafts: support-force control apparatus; transportation and lift systems; measurement instruments; energy storage and infrastructure; information and winding-register devices; time-rate metrology; and completion-scheduled computation). Each companion states its own relationship to this disclosure and remains self-contained as to claim-essential matter. Nothing in this specification depends on any companion for the meaning of a claim term.

FIELD

Engineered materials, resonant structures, modules, and manufacturing methods for **angular-momentum drive devices**—devices that apply an oscillating drive at the precession resonance of an aligned internal-circulation population of a solid medium and whose commanded output, whatever it proves to be, is taken from a stored empirical response map. More particularly: (i) **coherence media** as manufactured articles specified by a phase-storage property box under stated measurement conditions and by a size–frequency–saturation design rule, including substituted iron-garnet grades, cryogenic grades, structured (graded, layered, composite) media, and a separately claimed prophetic paramagnetic-resonance class; (ii) **phase-coherent cavity-array drive devices** distributing one locked source across N media with per-element feed-phase, amplitude, and alignment-field trim closed on a per-element precession-phase observable, and the method of acquiring and holding array lock; (iii) **travel-guide feeds** maintaining

resonant illumination of a medium in externally driven vertical travel; (iv) **stress-isolating, artifact-nulled mounts**; (v) **sealed drive modules** with split thermal paths, calibration memory, validity states, and enumerated interfaces; (vi) **tiled media assemblies** from gram to tonne class; and (vii) the **methods of making, factory-qualifying, and operating** each, including a self-contained factory validity-gate protocol with discriminating controls whose outputs are shipped with every module.

BACKGROUND

Extant engineering of resonant magnetic insulators optimizes for microwave filters and oscillators: small-signal linewidth, temperature stability of the resonance, and spurious-mode suppression in sub-millimetre spheres. The device class of this disclosure optimizes a different figure of merit—the total coherent phase that can be accumulated by the largest alignable population within one coherence interval, $\text{FoM} = f_{\text{align}} \times (v_{\text{drive}} \times \tau_{\text{coh}}) \times N_{\text{pop}}$ (dimensionless: aligned fraction $\times$ drive cycles per coherence interval $\times$ number of net aligned moments; Definition 8)—under simultaneous constraints that the filter art does not face: full- volume insulation for drive penetration, a single uniform precession mode in an element as large as the design rule permits, coherence retained at the operating drive amplitude rather than at small signal, thermal manageability of the medium itself at pulsed duty, mechanical mounting without magnetoelastic detuning, and manufacturing qualification that discriminates a genuine drive-synchronous force response from the ordinary magnetic-gradient, thermal, and servo artifacts that accompany any resonantly driven magnetic body on a force sensor. No published media specification, array architecture, mount, module, or factory test addresses this figure of merit or these constraints together.

Whether a resonantly driven aligned population produces any change in the support force of the body that carries it is the subject of the applicant's separate research program and is treated throughout this specification as a quantity to be **measured** by the factory qualification disclosed here—never as a presumed result. The media, arrays, mounts, modules, and assemblies disclosed are useful as manufactured articles and drive apparatus whether the measured force response of a given unit is nonzero or null: a null-qualified unit is a complete and useful output of the qualification method (a null reference, Definition 20).

Applicant's published exhibit corpus (nonessential background). Applicant maintains a public, commit-dated exhibit series presenting the underlying research program in didactic depth (the Normal Realism / POAMS exhibits, <https://cfmilam.github.io/nr-poams-exhibits/>). Those pages are applicant's own prior publications and are cited as nonessential background teaching only: no material essential to enablement, written description, or the claims is incorporated from them, and this specification is complete and self-contained.

SUMMARY

Six product families and their methods of making, qualifying, and operating are disclosed.

(1) **Coherence media (FIGS. 1–3).** Iron-garnet single-crystal bodies $R_3Fe_5O_{12}$ with Y and non-fast-relaxing R-site substituents (La, Lu, Bi, and Gd with care) and Fe-site Ga or Al substitution, specified by a property box (710) under stated measurement conditions—precession- resonance linewidth $\Delta H \leq 10$ Oe (preferred ≤ 1 Oe) at the operating frequency, resistivity $\geq 10^{10} \Omega \cdot \text{cm}$ at 295 K, saturation $4\pi M_s$ 200–2000 G at 295 K (application-selected), a large- signal coherence parameter, and, for the reference sphere geometry, diameter 2–15 mm and a polished surface class—together with a **size–frequency–saturation design rule** (FIG. 3) that co-selects element diameter d , saturation $4\pi M_s$, and drive frequency f so that the element is saturated and supports a single uniform precession mode: $f \geq (2.8 \text{ MHz/Oe}) \cdot 4\pi M_s/3$ and $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$. Lower-saturation grades are the technical means of realizing large elements at lower frequency; the rule is why the saturation window spans 200–2000 G. Cryogenic (77 K) grades carry a purity rider. Structured media—graded, layered, and composite bodies with named buildable species—meet the same property box. Dilute-paramagnetic single crystals (Cr:Al₂O₃ ruby class) are disclosed as a **separately claimed, prophetic paramagnetic-resonance medium class** with an explicit polarization condition; they do not possess a spontaneous net moment and are outside the ferrimagnetic property box. The named materials are references, not bounds: media developed hereafter that satisfy the property box and design rule are within the envelope. A method of making the medium (growth, orientation, forming, polishing, annealing, characterization, grading) is disclosed.

(2) **Phase-coherent cavity-array drive devices (FIGS. 4–5).** One locked source feeding an N-way corporate tree with per-branch phase and amplitude trim, N cavities each containing a medium, and—because two degrees of precession phase corresponds to about 20 kHz of detuning at a 1.12 MHz linewidth (≈ 0.007 Oe of alignment field)—a **per-element alignment-field trim actuator** driven by a per-element lock closed on a **per-element precession-phase observable** (the phase of the coupling-probe signal relative to the master reference). Phase coherence is defined by that observable, not by the feed network. Isolation ≥ 20 dB, or a deliberately coupled chain operated at its band-edge mode, and common crystallographic orientation are disclosed. The **method of operating** the array is a lock-acquisition sequence: master lock $\rightarrow$ per-element bias trim $\rightarrow$ per-element dither lock $\rightarrow$ feed-phase trim on the probe phase $\rightarrow$ thermal tracking $\rightarrow$ array response read only from a measured array allocation map. Additivity across elements is not presumed; it is an output of qualification.

(3) **Travel-guide feed (FIG. 6).** A slotted metallic traveling-wave guide with matched termination illuminating a dielectric travel tube through distributed slot or helix couplers, an alignment field spanning the travel to a stated uniformity or a tracking bias trim along the travel, and position sensing, so that a medium in **externally driven vertical travel** remains resonantly illuminated and locked. The motive cause of the travel is external to this disclosure.

(4) **Stress-isolating, artifact-nulled mount (FIG. 7).** A low-loss dielectric post with a strain-decoupling throat and minimal-area bond such that the stress transmitted to the medium produces a magnetoelastic resonance shift smaller than the medium linewidth ($3\lambda\sigma/M_s \approx 0.43$ Oe per MPa for yttrium iron garnet), with magnetostrictive-force nulling at the support interface as a secondary function; sapphire or alumina preferred for creep and thermal reasons.

(5) **Sealed drive module (FIGS. 8–9).** Housing, cavity and mount, bias magnet with trim windings, drive chain, controller and sense, shield partition, non-magnetic fasteners, **two thermal paths**—an electronics path to the baseplate rated in watts and a medium path through a high-conductivity post rated in milliwatts to about 100 mW with stated thermal resistance—a calibration memory holding the medium provenance, lock curve, force-offset transfer map, thermal drift map, validity envelope, and safe-off parameters, runtime witnesses, validity states, and an enumerated interface set (mounting pattern, DC power, RF ports, data/control bus, thermal interface). A module is shipped as a **qualified module** or as a **null reference** according to its map.

(6) **Tiled media assemblies (FIG. 2, 790).** Pluralities of media integrated in a common matrix or frame with individual (module) coupling structures and a common alignment-field bus; gram through tonne class stated as conditional prophetic scale embodiments realized only as tiled elements each within the design rule, driven through the array architecture of (2).

Factory qualification (FIGS. 10–11). A self-contained validity-gate protocol E1–E9 with pass criteria, including discriminating controls against the artifacts that pass naive gates: full-system field reversal compared with bias-only reversal; gradient scan and gradient-nulled or antiparallel-pair mounting; a **magnetically equivalent moment-modulation dummy** rather than a dummy mass; **heating-matched detuning**; time-class separation by drive modulation above the thermal corner; and distance scaling. Every module ships with a record of stated schema.

Layered scope and three-tier architecture (FIG. 12). L0 raw observables (linewidth, probe phase, raw synchronous force offset) → L1 qualified medium class and validated maps → L2 operation of arrays, modules, and assemblies from stored maps under validity states with safe-off. Tier 1 products and operating methods lead; Tier 2 factory qualification produces the stored maps; Tier 3 validity states, recalibration, and safe-off govern deployed operation.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 Figure-of-merit map across composition space (schematic): the claimed ferrimagnetic property box (710) with garnet grades (712, 714, 716); the paramagnetic-resonance class (718, 792) and compensated-order silicates (719) shown outside the box. FIG. 2 Medium geometries and products: single sphere (720) with surface class (722); sphere array (724); layered/graded/composite structured media (726) with named species (727, 728); tiled assembly (790) on an alignment-field bus (791). FIG. 3 Size–frequency–saturation design-rule chart (700): saturation floor (702), uniformity ceiling (704), allowed region (706), reference operating points (708). FIG. 4 Phase-coherent N-element cavity-array drive device: locked source (730) with master absorption lock (731), phase-reference distribution (733), N-way tree (732), per-branch ϕ /att trim (734), per-element alignment-field trim actuator (735), cavities with media (736), per-element coupling probe and phase detector (737), coupled-chain alternative (738), array controller (739). FIG. 5 Per-element control loop and array lock-acquisition sequence (method of operating). FIG. 6 Travel-guide feed: slotted traveling-wave guide (740), dielectric travel tube (741), distributed couplers (742), matched termination (743), medium in externally driven travel (744), tracking

bias trim (745), alignment field spanning travel (746), position sensor (747). FIG. 7 Stress-isolating, artifact-nulled mount: throat (750), low-loss post (752), high-conductivity post option (753), support interface (754), minimal-area bond (756), with the magnetoelastic budget. FIG. 8 Sealed drive module cross-section: housing (760), bias magnet and trim windings (762), drive chain (764), fasteners (766), controller and sense (768), shield partition (769), electronics thermal path (770), medium thermal path (771), calibration memory (772), runtime witnesses (773), interface set (774), safe-off (775). FIG. 9 Split thermal paths with conductance arithmetic (770, 771). FIG. 10 Factory qualification jig (780) and validity-gate flow E1–E9 (786) producing the ship record (784) with thermal drift map (782). FIG. 11 Discriminating controls: gradient trim coils and scan (781), moment-modulation dummy (783), heating-matched detune load (785), antiparallel pair (787), reaction frame and distance-scaling stage (788), time-class separation. FIG. 12 Three-tier architecture (796), layered scope L0–L2 (794), and validity-state machine (798).

DEFINITIONS (the applicant acts as his own lexicographer; these definitions govern the claims)

Portfolio operating terminology. "POAMS-native" and "direct-mode" identify the prophetic operating tier that applies the disclosed POAMS bookkeeping directly rather than requiring a preexisting nonzero empirical response map. "Commanded" and "selectable" identify controller- addressable input states and do not promise a nonzero output. "Independently validated" means reproduced under the disclosed qualification protocol against a separate reference; "empirically established" means recorded with uncertainty and provenance in a stored response map. Extant-framework labels appearing in quotation marks or introduced by "extant label:" identify measured phenomena by their customary names; their use commits neither the specification nor the claims to any entity ontology historically associated with those names. Every measured claim quantity is fixed by a recited instrument reading. Hypothesis parameters are identified expressly and are not presumed to have independently measured values. Constants used throughout: $\gamma/2\pi = 2.8 \text{ MHz/Oe}$; $c = 2.998 \times 10^8 \text{ m/s}$; $\epsilon_r \approx 15$ for yttrium iron garnet; $\mu_B = 9.274 \times 10^{-21} \text{ erg/G}$.

1. **Internal circulation / internal angular-momentum population** — the ensemble of constituent angular momenta of a solid medium; the quantity the extant literature labels "electron spin" and measures in aggregate as magnetization (standard magnetometry).
2. **Net uncompensated internal circulation** — the condition that the population is spontaneously ordered such that, at the operating temperature and under the alignment field, it presents a nonzero net aligned moment measurable as a saturation magnetization $4\pi M_s$ of at least the recited value (extant labels: ferrimagnetic or ferromagnetic order). Compensated order (extant label: antiferromagnetic) and purely thermal polarization of a dilute paramagnet do not satisfy this condition; the latter is addressed separately by Definition 12.
3. **Coherence medium / ordered medium** — a body possessing net uncompensated internal circulation and satisfying the property box of Definition 9 under the recited measurement

conditions. Reference medium: single-crystal yttrium iron garnet ($\text{Y}_3\text{Fe}_5\text{O}_{12}$, "YIG"; $4\pi M_s \approx 1750$ G at 295 K, $M_s \approx 139$ emu/cm³, density 5.17 g/cm³) and its Ga-, Al-, La-, Lu-, Bi-, and Gd-substituted grades.

4. **Alignment field; alignment sense; alignment-field trim** — the static bias field (oersted or tesla) applied to establish a measured alignment state of the population; "sense" is its polarity; "trim" is a commanded small change of its magnitude at one element (trim winding, shim, or temperature of a permanent-magnet element), resolution stated in oersted. A sphere is saturated when the applied field is at least $4\pi M_s/3$.
5. **Precession resonance; linewidth ΔH** — the absorption resonance of the aligned population under the alignment field (extant label: ferromagnetic resonance, FMR), located operationally as the absorption extremum to which a drive servo locks; for a sphere the resonance frequency is $f = (\gamma/2\pi) \cdot H_{\text{applied}}$. The linewidth ΔH is the full width at half maximum of the absorption in field units, measured at a stated frequency and temperature on the finished body; the equivalent frequency width is $\Delta f = (\gamma/2\pi) \cdot \Delta H$.
6. **Population coherence time τ** — the decay time of the coherent material precession, $\tau \approx 2/(\gamma\Delta H)$ with γ in rad/(s·Oe), or a time-domain pulsed measurement (extant label: T_2^*). $\Delta H = 0.4$ Oe gives $\tau \approx 0.284$ μs ; $\Delta H = 10$ Oe gives 11 ns.
7. **Drive turn rate v ; sufficient phase coherence** — v is the drive frequency in cycles per second. A medium has sufficient phase coherence under a recited drive when $\tau \cdot v \geq 1$ (at least one full turn of drive phase within one coherence interval). $\Delta H = 0.4$ Oe at 6 GHz gives $\tau \cdot v \approx 1700$; at 0.70 GHz, ≈ 200 ; $\Delta H = 10$ Oe at 6 GHz gives ≈ 68 .
8. **Figure of merit (FoM); f_{align} ; N_{pop}** — $\text{FoM} = f_{\text{align}} \times (v \cdot \tau) \times N_{\text{pop}}$, dimensionless. f_{align} is the aligned fraction M/M_s under the recited alignment field (1 at saturation). N_{pop} is the number of net aligned moments in the body, $N_{\text{pop}} = M_s \cdot V / \mu_B$ with M_s in emu/cm³ and V the body volume (a 10 mm YIG sphere: $V = 0.5236$ cm³, $m = 72.8$ emu, $N_{\text{pop}} \approx 7.9 \times 10^{21}$; a 10 mm garnet sphere of $4\pi M_s = 600$ G: $m = 25$ emu, $N_{\text{pop}} \approx 2.7 \times 10^{21}$). The FoM is a stated selection criterion for media; it is not a claim of achieved device performance, and the relation between FoM and any measured force response is established only by the qualification of this disclosure.
9. **Property box** — the set of measured limits a medium must satisfy, each with its measurement condition: linewidth ΔH at the operating frequency (or at a stated reference frequency) and at the grade temperature (295 K or 77 K) on the finished polished body; resistivity ρ at 295 K by two-terminal guarded measurement; saturation $4\pi M_s$ at 295 K by vibrating-sample magnetometry; surface class as the arithmetic-mean roughness R_a and maximum pit diameter after final polish; the large-signal coherence parameter of Definition 10 at the operating frequency; and, for spheres, diameter and sphericity.
10. **Large-signal coherence parameter** — either (a) the threshold drive field h_{crit} at the operating frequency and temperature at which the uniform precession mode ceases to grow linearly with drive (extant label: spin-wave instability / premature saturation threshold), or (b) the linewidth

measured at the operating drive amplitude; both are measured on the finished body in its cavity and recorded; numerical limits are learned in qualification.

11. **Size–frequency–saturation design rule** — the co-selection of element diameter d , saturation $4\pi Ms$, and drive frequency f such that (i) the body is saturated at resonance, $f \geq (2.8 \text{ MHz/Oe}) \cdot 4\pi Ms/3$, and (ii) the body supports a single uniform precession mode, $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$ (equivalently $k_0 \cdot d \cdot \sqrt{\epsilon_r} \leq 2\pi/10$). See FIG. 3 and the design-rule section.
12. **Paramagnetic-resonance coherence medium (prophetic class)** — a dilute-paramagnetic single crystal (reference: $\text{Cr}^{3+}:\text{Al}_2\text{O}_3$, "ruby") having a paramagnetic-resonance linewidth at the operating frequency not exceeding a stated value, and operated under a stated polarization condition—thermal polarization $p = \tanh(h\nu/2kT)$ at a stated temperature and frequency, or a pumped polarization exceeding a stated value—such that the aligned moment density $p \cdot n_{\text{spin}}$ exceeds a stated floor. This class has no net uncompensated internal circulation and lies outside the property box of Definition 9.
13. **Angular-momentum drive device** — a device comprising a coherence medium, an alignment-field source, a coupling structure, and a drive chain locked to the medium's precession resonance, operated to produce drive-synchronous coherent phase advance of the aligned population, whose commanded outputs, if any, are taken from a stored empirical response map. No output sign, magnitude, additivity, or scaling is presumed by this definition.
14. **Per-element precession-phase observable** — the phase of the signal at an element's coupling probe (or of the reflected/transmitted wave at its port), at the drive frequency, relative to the master phase reference distributed to that element, after subtraction of the recorded feed electrical length. For a resonantly driven population the observable lags the local drive by $\arctan(2\delta/\Delta f)$ where δ is the detuning; 2° corresponds to $\delta = \frac{1}{2} \cdot \Delta f \cdot \tan 2^\circ \approx 19.6 \text{ kHz}$ at $\Delta f = 1.12 \text{ MHz}$, i.e. $\approx 0.007 \text{ Oe}$ of alignment field.
15. **Phase-coherent array** — an array in which every element's precession-phase observable lies within a stated tolerance (target class $\pm 2^\circ$) of the array's commanded phase pattern, as measured, while every element is locked to the common drive frequency.
16. **Force offset (ΔF)** — the change in the **support force** of a defined body—the medium, or the mount, module, or assembly rigidly bearing it—read by a force sensor **along the support axis** (the local vertical of the qualification jig unless otherwise stated), synchronized with a commanded change of drive state and referred to the drive-off baseline: $\Delta F_s = \text{mean}(F_{\text{ON}}) - \text{mean}(F_{\text{OFF}})$ at alignment sense s , or the in-phase and quadrature components at a modulation frequency, with statistical uncertainty. ΔF may be zero.
17. **Force-offset transfer map (transfer function)** — the stored empirical relation from a defined input—the drive-state vector of Definition 23 (at least absorbed drive power, duty, detuning, alignment-field magnitude and sense, medium temperature)—to the force offset, with uncertainty, hysteresis, and time constants, established on the qualification jig against a separate force reference. The map admits zero at every input; no functional form is presumed.

18. **Channel sign; differential channel** — a *differential channel* is the synchronously computed difference of the force reading between two paired states (drive ON/OFF; alignment sense +/-; active medium/reference dummy; gradient +G/-G), together with its noise floor R (the standard deviation of the differential channel over the stated averaging interval, determined from drive-off data). The *channel sign* is the measured sign of the sense-odd component $(\Delta F_+ - \Delta F_-)/2$ of a unit's force offset at a validated operating point. It is a recorded datum of the unit; no sign is presumed, and its physical interpretation (the applicant's channel-partition hypothesis) is not relied on for any utility here.
19. **Stored empirical response map; provenance record** — the data structure held in a module's calibration memory: medium identity and grade; lock curve (resonance frequency versus alignment field and temperature, linewidth, loaded Q); force-offset transfer map; thermal drift map (resonance and force offset versus temperature); the identity of every validated operating point and the gates it passed; noise floor R; provenance (jig, force reference, date, operator, raw-data identifier, fit method, replication record). A map may contain only null operating points.
20. **Qualified medium class; qualified module; null reference** — a *qualified medium class* is a production lot or grade whose sampled bodies satisfy the property box under the stated measurement conditions and whose provenance is recorded. A *qualified module* is a unit (or class represented by qualified units) whose map holds at least one operating point passing gates E1–E9 with a nonzero force offset, its uncertainty, provenance, validity envelope, and safe-off parameters. A unit whose map holds only null operating points is a *null reference*: it may be operated as a drive device, reports L0 and L1 quantities, and has no enabled force-setpoint mode. Neither status presumes any response beyond the validated points.
21. **Validity envelope; runtime limits; validity states; safe-off** — the region of drive and environmental state over which the maps were established (including permissible interpolation) and the runtime limits (medium and mount temperature, orientation, support- axis field gradient, vibration, absorbed power and duty, lock error, probe-phase error, drift rate, zero-check interval) beyond which the controller reports *out-of-envelope*. Validity states: *valid*, *degraded*, *out-of-envelope*, *invalid* (map or gate check failed; recalibration flagged), *safe-off*. Safe-off: oscillating drive off; alignment field to a stored safe magnitude and sense (or unchanged in a fixed-bias unit); last valid record preserved; recalibration flag set.
22. **Array allocation map** — for an array or tiled assembly, the measured relation between per-element drive states and the assembly-level force offset, established under sequential, in-phase, phase-scrambled, and inactive-element qualification states with cross-talk terms. The sum of per-element offsets is not presumed.
23. **Drive-state vector** — the commanded and independently measured operating variables for one observation: alignment-field magnitude and sense at each element; drive frequency, per-element feed phase, incident, reflected, and absorbed power; pulse width, repetition rate, duty, modulation frequency; resonance mode; probe-phase observables; medium, mount, and ambient temperature; support-axis field gradient; orientation; time.

24. **Heating-matched (absorbed-energy-matched) control** — a control state (detuned drive plus a resistive or radiative load in the module, or a far-detuned drive at raised power) adjusted so that the medium and module temperature histories match the resonant state within a stated tolerance, verified by the medium temperature witness; equality of source power alone is not sufficient.
25. **Externally driven vertical travel** — motion of a medium along the travel guide produced by an actuator, hoist, stage, or vehicle external to this disclosure; the guide of FIG. 6 feeds and locks the medium during that travel and is not the motive cause of it.

MEDIA DESIGN RULES (FIGS. 1–3)

Design rule (700). The premise of a single sharp precession resonance in a spherical body (uniform internal-field response; extant label: demagnetizing factor $N = 1/3$) holds in the magnetostatic limit, when the body is small compared with the dielectric wavelength inside it. The applicant states the limit as $k_0 \cdot d \cdot \sqrt{\epsilon_r} \leq 2\pi/10$, i.e. $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$ (uniformity ceiling, 704); with $\epsilon_r \approx 15$ the maximum diameter is 1.29 mm at 6 GHz, 4.75 mm at 1.63 GHz, 7.74 mm at 1.0 GHz, 11.1 mm at 0.70 GHz, 20.9 mm at 0.37 GHz, and 40.7 mm at 0.19 GHz. A 10 mm sphere driven at 6 GHz has $k_0 \cdot d \cdot \sqrt{\epsilon_r} \approx 4.9$ and is a multi-mode dielectric-loaded magnetostatic resonator; it is operable as a drive body whose locked mode is recorded, but it is outside the single-mode design region and is not used as a reference operating point here. Saturation of a sphere at resonance requires an applied field of at least $4\pi Ms/3$, so the resonance frequency cannot lie below $f_{\min} = (2.8 \text{ MHz/Oe}) \cdot 4\pi Ms/3$ (saturation floor, 702): 1.63 GHz for pure YIG (1750 G), 0.75 GHz at 800 G, 0.56 GHz at 600 G, 0.37 GHz at 400 G, 0.19 GHz at 200 G. The allowed region (706) lies above the floor and below the ceiling. Consequences: a pure-YIG sphere larger than 4.75 mm cannot be both saturated and single-mode; 10–15 mm elements are realized only with low-saturation grades (≈ 800 G) at ≈ 1 GHz. **This is the technical reason for the 200–2000 G saturation window:** the saturation grade is the design variable that sets the largest single-mode element for a chosen drive frequency. Reference operating points (708): pure YIG, 1.2 mm, 6.0 GHz; pure YIG, 4.5 mm, 1.7 GHz; Ga:YIG 600 G, 10 mm, 0.70 GHz ($d \cdot f \cdot \sqrt{\epsilon_r} = 2.71 \times 10^7 \text{ m/s} \leq c/10 = 3.0 \times 10^7 \text{ m/s}$; applied field $250 \text{ Oe} \geq 4\pi Ms/3 = 200 \text{ Oe}$).

Substitution rationale. Fe-site Ga or Al substitution lowers $4\pi Ms$; its engineering purposes are (i) a lower saturation floor and hence a lower drive frequency, (ii) a larger single-mode element at that frequency, (iii) a higher spin-wave-instability threshold, and (iv) a smaller stray dipole field in arrays (a 10 mm sphere of 600 G presents 0.026 Oe at 12.5 cm versus 0.075 Oe for pure YIG). R-site substituents are restricted to the non-fast-relaxing set—La, Lu, Bi, and Gd (the latter with attention to its magnetization compensation near 290 K)—because Tb, Dy, Ho, Er, and Yb broaden the linewidth to tens or hundreds of oersted. At equal volume a low-saturation grade has fewer net moments (600 G: 0.34 of pure YIG) and a lower FoM; the FoM gain arises because the design rule permits the larger body. Table 1 (PROPHETIC EXAMPLES, P1) evaluates the FoM at the reference operating points.

Property box (710) and measurement conditions. $\Delta H \leq 10$ Oe (preferred ≤ 1 Oe) at the operating frequency on the finished polished body at the grade temperature; $\rho \geq 10^{10} \Omega \cdot \text{cm}$ at 295 K; $4\pi M_s$ 200–2000 G at 295 K (pure YIG reaches ≈ 2460 G at 77 K; the window is a 295 K specification); surface class $R_a \leq 0.5 \mu\text{m}$ and pit diameter $\leq 0.5 \mu\text{m}$ for the ≤ 10 Oe grade, $R_a \leq 0.1 \mu\text{m}$ for the preferred ≤ 1 Oe grade (surface pits broaden the linewidth by two-magnon scattering); large-signal coherence parameter recorded at the operating frequency. Because the first-order spin-wave instability requires the half-frequency spin-wave band to be reachable, and for a pure-YIG sphere at 6 GHz the band bottom (γH_i with $H_i = 2143 - 583 = 1560$ Oe $\rightarrow 4.37$ GHz) lies above $\omega/2 = 3$ GHz, the first-order process is absent there; the second-order (premature-saturation) process is always present and sets h_{crit} , which Ga substitution raises. Resistivity doubles as a purity proxy: Fe^{2+} (and Si, Pb) lower ρ and, with ppm-level fast-relaxing rare-earth impurities, produce linewidth peaks in the 20–100 K range; the **77 K grade (cryogenic grade)** therefore carries the rider $\rho \geq 10^{10} \Omega \cdot \text{cm}$ and fast-relaxing rare-earth content below 10 ppm, with $\Delta H \leq 1$ Oe measured at 77 K at the operating frequency.

Structured media (726). Graded, layered, and composite bodies meeting the property box as a composite: species (727) a stack of liquid-phase-epitaxy garnet films of graded Ga content on gadolinium gallium garnet substrates, each film satisfying ΔH and ρ , bonded with low-loss adhesive; species (728) garnet spheres each within the design rule embedded in a low-loss dielectric matrix (PTFE, polyethylene, or alumina-loaded resin, $\tan \delta \leq 10^{-3}$) at a fill factor and pitch chosen so that the inter-sphere dipolar cross-field remains below the linewidth. The composite is characterized by the property box measured on the composite body.

Paramagnetic-resonance class (718, 792; prophetic). Ruby ($\text{Cr}^{3+}:\text{Al}_2\text{O}_3$) has a paramagnetic-resonance line of order 4–7 Oe but no spontaneous moment: at 6 GHz and 77 K its thermal polarization is $\tanh(h\nu/2kT) \approx 1.9 \times 10^{-3}$, so 0.05 at.% Cr ($2.35 \times 10^{19} \text{cm}^{-3}$) yields $\approx 4.4 \times 10^{16}$ aligned moments per cm^3 , about five orders below the $2 \times 10^{22} \text{cm}^{-3}$ of ordered garnet. It is therefore disclosed and claimed separately, with the polarization condition of Definition 12, and is useful where its linewidth stability, radiation hardness, or optical access are required and the population floor can be met by pumping or concentration. Olivine-class silicates (719) are compensated (fayalite orders antiferromagnetically near 65 K) or paramagnetic and are disclosed only as a conditional alternative of the same paramagnetic class; neither is within the ferrimagnetic property box. The named materials are references; media developed hereafter that satisfy the property box and design rule—synthetic, composite, or structured—are within the envelope.

METHOD OF MAKING THE MEDIUM

(a) Grow a single-crystal boule of the selected garnet composition (flux growth in $\text{PbO-B}_2\text{O}_3$ or Czochralski growth), the Ga or Al content selected from the design rule for the intended element size and frequency. (b) Orient the boule by X-ray diffraction and mark a common crystallographic axis ($\langle 111 \rangle$ preferred) on every element so that the anisotropy-field contribution to the resonance field is common across an array ($|K_1|/M_s \approx 44$ Oe for YIG at 295 K; the $\langle 100 \rangle$ – $\langle 111 \rangle$ resonance-field spread for a sphere is

≈ 146 Oe). (c) Core, cube, and tumble-grind to a sphere of the selected diameter; polish to the surface class of the grade ($R_a \leq 0.1 \mu\text{m}$ for the ≤ 1 Oe grade). (d) Anneal in oxygen ($\approx 1000\text{--}1400^\circ\text{C}$) to convert Fe^{2+} and restore resistivity. (e) Characterize each body or a sampling plan: ΔH at the operating frequency (and at 77 K for the cryogenic grade) in a reference cavity; $4\pi M_s$ at 295 K; ρ at 295 K; surface class by profilometry and microscopy; large-signal parameter in the reference cavity; sphericity; orientation mark. (f) Grade the lot against the property box, assign the qualified medium class identifier, and write the medium provenance record. A body failing the box is not a coherence medium of this disclosure.

PRODUCT ARCHITECTURES

M1 — Medium article (720, 722, 724, 726–729). A finished, graded, oriented body of a qualified medium class with its provenance record: the reference sphere 2–15 mm (720) with surface class (722); sphere arrays (724); structured species (727, 728); and the recorded large-signal parameter (729). Larger boules, centimetre-to-decimetre elements, and tiled assemblies (790) aggregate unit media each within the design rule.

M2 — Phase-coherent cavity-array drive device (FIG. 4). A locked source (730) is servo-locked to the precession resonance by a master absorption lock (731) on a reference element or on the array's summed absorption. A phase-reference distribution (733) carries the master phase to every element. An N-way corporate tree (732) feeds per-branch voltage-controlled phase shifters and attenuators (734). Each element comprises a cavity containing a medium (736), a coupling probe with a phase detector referenced to the distributed master phase (737), and an alignment-field trim actuator (735)—a trim winding on the element's bias magnet (resolution ≤ 0.005 Oe, i.e. below the 0.007 Oe that corresponds to 2° at a 1.12 MHz linewidth; 28 ppm of a 250 Oe bias, 3.3 ppm of 2143 Oe), or a thermal trim of a permanent-magnet element. An array controller (739) closes, for every element, (i) a per-element dither lock that modulates the element's trim actuator and holds the element's absorption extremum at the common drive frequency, and (ii) a feed-phase loop that trims the branch phase shifter (734) so that the element's precession-phase observable equals its commanded value. Inter-cavity isolation ≥ 20 dB limits the worst-case phase pull from a neighbor's leakage to about 5.7° , which the per-element loop removes as a static offset (≥ 30 dB $\rightarrow \leq 1.8^\circ$ open-loop). Spheres are mounted with their marked crystallographic axis common, and the frame pitch is chosen so that the dipolar cross-field is below the linewidth (0.026 Oe at 12.5 cm for the 600 G, 10 mm element). Alternative (738): a deliberately coupled cavity chain operated at its band-edge mode, where the relative element phases are fixed by the mode index; per-branch trim is then replaced by tuning the chain to the band edge, and the per-element bias trim remains. Conformal arrays on frames (platform geometry) carry a common alignment-field bus (791): coil sets or a permanent-magnet lattice with per-element trim windings. The array response, if any, is read only from the array allocation map (Definition 22).

Method of operating the array (FIG. 5). (a) Energize the alignment-field bus to the mapped magnitude and sense; verify each element's bias witness. (b) Acquire the master lock (731) on the reference element. (c) For each element, sweep the trim actuator (735) to locate the element's absorption extremum at the

common frequency and engage its dither lock. (d) Read each element's precession-phase observable (737) and trim the branch phase shifter (734) to the commanded phase pattern; verify all elements within tolerance (target $\pm 2^\circ$). (e) Engage thermal tracking: the dither lock follows the resonance drift with temperature (0.1–0.5 MHz/K class), and the controller reports out-of-envelope if a trim actuator approaches its range limit. (f) Apply the commanded drive envelope (pulse, burst, or CW within the duty bound) and log the drive-state vector. (g) Read any assembly-level output only from the stored array allocation map; hold safe-off if the map is null or the state is outside its envelope. (h) On loss of lock at any element, drop that element to inactive, re-acquire, and re-verify phase.

M3 — Travel-guide feed (FIG. 6). A slotted metallic traveling-wave guide (740)—a rectangular or circular waveguide with a longitudinal array of slot or helix couplers (742) and a matched termination (743) so that no standing-wave null forms along the travel—runs alongside a dielectric travel tube (741, PTFE, polycarbonate, or alumina) within which a medium (744) is moved by an externally driven vertical travel. The alignment field (746) spans the travel with a uniformity of one linewidth over the travel length (0.4 Oe of 250 Oe, i.e. 1.6×10^{-3} , for the 600 G reference element; 2×10^{-4} for a 0.4 Oe line at 2143 Oe), or a tracking bias trim (745) distributed along the travel and commanded from the position sensor (747, optical) holds the medium on resonance as it moves. Doppler shift at 1 m/s is ≈ 20 Hz at 6 GHz—negligible against the linewidth. The medium's lock and probe-phase observable are read through the coupler nearest the medium, selected by position. The guide is a feed and lock architecture; it is not the motive cause of the travel.

M4 — Stress-isolating, artifact-nulled mount (FIG. 7). The medium (720) is carried on a low-loss dielectric post (752) through a strain-decoupling throat (750) and a minimal-area bond (756), to a support interface (754) that transmits the medium's support force to the force sensor or load-bearing structure. The primary function is magnetoelastic isolation: mount stress σ shifts the resonance by $\approx 3\lambda\sigma/M_s$, which for YIG ($\lambda_s \approx -2 \times 10^{-6}$, $M_s = 139$ emu/cm³) is 0.43 Oe per MPa—a full narrow linewidth—while adhesive cure shrinkage and differential expansion (PTFE $\alpha \approx 10^{-4}$ /K) reach megapascals. The throat is dimensioned so that the stress transmitted to the medium produces a shift smaller than the medium linewidth over the operating temperature range. The secondary function is artifact nulling: a simply supported sphere transmits no steady force from its own magnetostrictive expansion (≈ 20 nm on 10 mm), and the throat keeps any clamped- geometry bending moment from reaching the sensor. Sapphire or alumina posts (753) are preferred over PTFE for creep, expansion, and thermal conductance (M5).

M5 — Sealed drive module (FIGS. 8–9). A drive-transparent structural housing (760; polycarbonate or PEEK), rated for the stated containment (vacuum or dry-gas fill, and fragment retention); a cavity with the mount of M4 (736/752); a bias magnet with trim windings (762); a drive chain (764: source or distributed feed input, modulator with duty bound, amplifier, directional coupler, incident/reflected/transmitted detectors); a controller and sense unit (768) with a shield partition (769) separating coil fields from sensor electronics; non-magnetic fasteners and adhesives (766); **two thermal paths**: an electronics and coil path (770) to the baseplate rated in watts, and a medium path (771) from the medium through the post to the baseplate, rated by its conductance—a 3 mm $\times$ 10 mm PTFE post conducts 1.8×10^{-4} W/K (≈ 5700 K/W) and can carry only about 1 mW at a 6 K rise, whereas a 6 mm $\times$

10 mm sapphire post conducts 0.11 W/K (8.8 K/W) and carries 100 mW at 0.9 K, which at 0.5 MHz/K is 0.44 MHz, below a 1.12 MHz linewidth; radiation and convection together add only $\approx 5 \times 10^{-3}$ W/K. The medium duty envelope is set so that $\Delta T \cdot (df/dT)$ stays below the linewidth or within the dither lock's tracking range. A calibration memory (772) holds the stored empirical response map (Definition 19); runtime witnesses (773) monitor medium and mount temperature, bias, lock error, probe phase, absorbed power and duty, support-axis gradient, vibration, orientation; the controller maintains the validity state and executes safe-off (775). The interface set (774) is enumerated: mounting pattern and datum; DC power; RF port(s) or distributed-feed port; data/control bus carrying the drive-state vector, validity state, and map identifier; thermal interface at the baseplate; optional support-path force-sensor port. The module ships as a qualified module or as a null reference.

M6 — Tiled media assembly (FIG. 2, 790). A plurality of media of M1 mechanically integrated in a common matrix or frame with individual coupling structures (modules of M5) and a common alignment-field bus (791), driven by the array device of M2 and qualified under the array allocation map. At 2.7 g per 10 mm element, 0.1 kg is 37 elements, 1 kg 369, 100 kg 3.7×10^4 , 1000 kg 3.7×10^5 (0.19 m³ of garnet alone). Because resonant absorption in a continuous large body is confined near the illuminated surface, kilogram-and-above assemblies are realized as tiled elements each within the design rule (or as structured media of species 727/728), never as a single bulk body; the bias volume, feed-tree depth, and thermal paths scale with element count. The mass tiers are conditional prophetic scale embodiments.

LAYERED SCOPE (FIG. 12, 794)

- **L0 — raw observables:** linewidth, $4\pi Ms$, ρ , surface class, and large-signal parameter of a body; the per-element probe phase and lock error of an array; the raw synchronous force offset ΔF of a unit on the jig with its noise floor R . Always reportable; ΔF may be zero.
- **L1 — qualified class and validated maps:** the qualified medium class; the lock curve, transfer map, drift map, and array allocation map of a unit, reportable inside the validity envelope. Gates E1–E9 stand between L0 and L1 for any nonzero force offset.
- **L2 — operation from maps:** the array, module, or assembly operated in the *valid* state from its stored maps; a null reference operates as a drive device with no force-setpoint mode.

QUALIFICATION, CALIBRATION, AND DEPLOYMENT (FIGS. 10–12)

Three tiers (796). Tier 1: the products of M1–M6 and their operating methods lead. Tier 2: factory qualification on the jig (780) produces the stored maps and classifies the unit. Tier 3: validity states, recalibration, and safe-off (798) govern deployed operation; the operator of a deployed unit does not repeat the gates.

Qualification jig (780). A reference balance or force transducer (noise floor R stated with its averaging interval; reference class 0.1 μN over 100 s) supporting the module through its support interface; a reaction frame (788) carrying the alignment-field source and feeds on a load path separate from the balance, with a reaction-force witness; a field fixture with reversible bias and a trim lattice; gradient trim

coils and a gradiometer (781) at the medium position; a moment-modulation dummy (783); a heating-matched detune load (785); an antiparallel pair fixture (787); a distance-scaling stage (788); temperature, vibration, pressure, and orientation witnesses; synchronized acquisition of the drive-state vector.

Factory validity gates E1–E9 (786), one paragraph each.

- **E1 — Repeatability.** The registered ON/OFF differential channel repeats across randomized blocks and across days within its stated uncertainty. Pass: block-to-block scatter consistent with R; no trend with block index.
- **E2 — Resonance contrast with heating match.** The response differs from a heating-matched detuned state (Definition 24: drive detuned ≥ 500 MHz, ≈ 179 Oe, with module dissipation and medium temperature history matched by the load 785) by the preregistered amount. Pass: $|\Delta F_{\text{res}} - \Delta F_{\text{detuned, matched}}| \geq 3R$ and the medium temperature witness matched within tolerance. A null at detune without heating match is not a pass.
- **E3 — Moment-equivalent material contrast.** The response is not reproduced by a magnetically equivalent dummy (783): a small coil or far-detuned garnet body of equal static moment m driven to produce the same $\delta m(t)$ waveform as the on-resonance M_z dip ($\theta = 1^\circ$ gives $\delta m/m \approx 1.5 \times 10^{-4}$), nor by a non-magnetic dummy mass. Pass: dummy channel within R; active channel $\geq 3R$ above it. A dummy mass alone is not a complete control.
- **E4 — Directional signature by full-system reversal.** The channel sign reverses when every field source (bias magnet or coil polarity **and** trim lattice) is reversed in a separate settled block, and the result is compared with a bias-only reversal; the difference between the two reversals is the recorded gradient-artifact asymmetry. Pass: full-system sense-odd component $\geq 3R$ with sense-even residual accounted for.
- **E5 — Gradient exclusion.** Map the support-axis gradient at the medium position; apply deliberate gradients $\pm G$ with the trim coils (781) and record ΔF versus G ; the intercept at $G = 0$ is the reportable offset and the slope is the Kelvin-force calibration (for $m = 73$ emu, 1 Oe/cm gives 0.73 mN static; the resonance-modulated part at $\theta = 1^\circ$ is 0.11 μN per Oe/cm; 1 K of heating gives 1.2 μN per Oe/cm; for the 25 emu reference element these are 0.25 mN, 0.038 μN , and 0.43 μN). Alternatively or additionally, mount two media with antiparallel bias in the same gradient (787): Kelvin terms cancel; any bias-sign-dependent response doubles or cancels, either being informative. Pass: residual gradient uncertainty below one third of the gradient that would imitate the smallest reportable offset.
- **E6 — Thermal exclusion with time-class separation.** Medium and mount temperature are witnessed; the drive is modulated at f_{mod} well above the thermal corner (thermal time constant of sphere plus mount is seconds; f_{mod} of 1–10 Hz) and the force channel is demodulated synchronously: a thermal artifact rolls off above $1/\tau_{\text{th}}$, a direct response follows the envelope. Pass: the demodulated channel's frequency dependence is inconsistent with the measured thermal corner, and the heating-matched detune of E2 shows no response.

- **E7 — Mechanical, electrical, and servo exclusion.** Vibration, strain, cable and feed forces, reaction-path force, bias-coil current forces, and stray-field coupling to the balance's own servo or magnetic parts are witnessed; the medium is repeated at two distances from the sensor mechanism (788), stray-field artifacts scaling $\approx 1/r^3$. Pass: no witness channel accounts for the response; distance dependence inconsistent with $1/r^3$.
- **E8 — Statistical validity.** $|\Delta F| \geq 3R$ over the stated averaging after correction for the preregistered search space and multiple comparisons; balance readability is not treated as standard deviation. Pass: stated.
- **E9 — Independent confirmation (class qualification).** A second jig, unit, run series, or laboratory reproduces the registered sign and response surface within uncertainty. Required to qualify a production class; a single unit lacking E9 ships as individually qualified only for the points that passed E1–E8, or as a null reference. Only an operating point passing E1–E9 (E1–E8 for an individually qualified unit) is a validated nonzero point of the map. A null at every point is a complete and useful qualification; the unit is a null reference and remains a functioning drive device.

Ship record (784) — schema. Module identifier and medium class identifier; medium provenance (composition, $4\pi Ms$, ρ , ΔH at frequency and temperature, surface class, large-signal parameter, orientation); lock curve (f versus H at stated temperatures, slope, linewidth, loaded Q); force-offset transfer map (input vector $\rightarrow \Delta F$ in newtons with sign and uncertainty, or null); channel sign at each validated point; thermal drift map (782: f and ΔF versus T ; lock tracking coefficient); noise floor R and averaging; gate results E1–E9 with values; validity envelope and runtime limits; safe-off parameters; qualification status (qualified module / null reference).

Empirical-map procedure. Acquire the randomized sequence of drive and control states across the intended envelope; fit the maps with a preregistered model class (lookup with interpolation or physics-agnostic regression) with covariance and time constants; classify each point as null, identified artifact, or unexplained drive-correlated residual; apply E1–E8 to any nonzero residual, E9 for a class; store; ship; update on runtime zero checks. What qualification does not presume: sign, magnitude, linear scaling with population or power, additivity across elements, or any performance beyond the validated points.

Array qualification. The assembled array or tiled assembly is qualified under sequential, in-phase, phase-scrambled, and inactive-element states with per-element and assembly-level force and reaction channels, producing the array allocation map (Definition 22).

Deployed operation and recalibration. A deployed module or array loads its maps, verifies provenance, runs a drive-off zero check, enters *valid* only if the witnesses lie within the runtime limits, operates from the maps, interleaves zero checks, transitions to *degraded*, *out-of-envelope*, *invalid*, or *safe-off* per Definition 21, and returns to a reduced re-qualification (drive-off and sense-reversed zero checks plus one validated point against a separate force reference) or to the full protocol on flag.

PROPHETIC EXAMPLES (prospective; every number follows from the stated constants and design rule; no result is represented as achieved)

P1 — Medium lot and FoM table. A lot of Ga-substituted YIG spheres, 10 mm, $4\pi M_s \approx 600$ G at 295 K, oriented $\langle 111 \rangle$, polished to $R_a \leq 0.1$ μm , annealed, with target linewidth ≤ 0.4 Oe at 295 K and ≤ 0.15 Oe at 77 K measured at the 0.70 GHz operating frequency (these targets are optimistic for Ga:YIG and require the finer surface class), $\rho \geq 10^{10}$ $\Omega\cdot\text{cm}$, fast-relaxing rare-earth content < 10 ppm, is expected to grade as a qualified medium class. Table 1 evaluates the FoM ($f_{\text{align}} = 1$) at design-rule-compliant operating points:

Species (295 K unless stated)	d (mm)	f (GHz)	$d \cdot f \cdot \sqrt{\epsilon_r}$ (10^7 m/s)	f_min (GHz)	ΔH (Oe)	$v \cdot \tau$	N_pop	FoM
pure YIG, 1750 G	1.2	6.0	2.79	1.63	0.4	1705	1.4×10^{19}	2.3×10^{22}
pure YIG, 1750 G	4.5	1.7	2.96	1.63	0.4	483	7.2×10^{20}	3.5×10^{23}
Ga:YIG, 600 G (P1)	10	0.70	2.71	0.56	0.4	199	2.7×10^{21}	5.4×10^{23}
Ga:YIG, 600 G, 77 K (P1)	10	0.70	2.71	0.56	0.15	531	2.7×10^{21}	1.4×10^{24}
pure YIG, 10 mm at 6 GHz	10	6.0	23.2 (outside rule)	1.63	—	—	—	not a valid entry
ruby 0.05 at.% Cr, 77 K, thermal (class 792)	10	6.0	—	—	≈ 5	136	2.3×10^{16}	3×10^{18}

The low-saturation large element leads the compliant entries by about $1.5\times$ at 295 K and $4\times$ at 77 K; the ruby entry shows the population floor the paramagnetic class must overcome by concentration or pumping.

P2 — Sixteen-element array. Sixteen P1 elements on a 0.5 m frame (4×4 , 16.7 cm pitch; dipolar cross-field ≈ 0.011 Oe at that pitch, below the 0.4 Oe linewidth), one 0.70 GHz source, bias 250 Oe from a permanent-magnet lattice with per-element trim windings of 0.005 Oe resolution (20 ppm), isolation ≥ 20 dB, per-element dither locks and probe-phase loops, are projected to reach a phase-coherent state with every probe-phase observable within $\pm 2^\circ$ of the commanded pattern. The array's force-offset behavior, if any, is to be read from its measured array allocation map; no additive or other scaling is projected.

P3 — Module qualification. A sealed module of one P1 element on a 6 mm sapphire post is expected to complete gates E1–E9 on a jig of noise floor $R = 0.1$ μN over 100 s, so that the smallest reportable offset is 0.3 μN . The gradient budget follows: 1 K of medium heating on the 25 emu element imitates 0.43 μN per Oe/cm, so E5 requires the residual support-axis gradient to be trimmed below ≈ 0.08 Oe/cm or the heating-matched detune of E2 to bound the thermal term; the resonance-modulated moment at $\theta = 1^\circ$ imitates 0.038 μN per Oe/cm. The outcome—qualified module or null reference—is not projected.

P4 — Thermal envelope. With the electronics path (770) carrying 5 W to the baseplate and the medium path (771) a 6 mm × 10 mm sapphire post (0.11 W/K), a medium average dissipation of 100 mW produces a 0.9 K rise and, at 0.5 MHz/K, a 0.44 MHz shift, tracked by the dither lock within the 1.12 MHz linewidth. The same dissipation through a 3 mm PTFE post would produce a rise of several hundred kelvin; PTFE posts are therefore limited to ≈ 1 mW medium dissipation.

P5 — Tiled assembly (conditional). A 1 kg assembly of 369 P1 elements populating a 20 × 19 frame segment on a common alignment-field bus, driven by a 369-way tree of the M2 architecture, is a conditional prophetic scale embodiment whose assembly-level behavior would be read only from its array allocation map.

POAMS-NATIVE MEDIA SCOPE

The YIG uniform-mode, size-frequency, linewidth, and thermal rules define enabled reference species; they are not universal limits on every POAMS coherence medium. Structured, extended, topological, and other whole-turn-supporting media remain prophetic species when they provide a measurable internal circulation, phase-coherent drive state, and committed winding response. Factory maps qualify production articles but do not replace the native angular-momentum organization the articles are built to support.

CLAIMS

1. A coherence medium for an angular-momentum drive device, comprising a single-crystal body having a net uncompensated internal circulation with a saturation magnetization $4\pi M_s$ of 200–2000 G measured at 295 K, a precession-resonance linewidth $\Delta H \leq 10$ Oe measured at a stated operating frequency and grade temperature on the finished body, a resistivity $\geq 10^{10} \Omega \cdot \text{cm}$ measured at 295 K, a recorded large-signal coherence parameter at the stated operating frequency, and a recorded medium provenance identifying a qualified medium class.
2. The medium of claim 1, comprising an iron garnet $R_3\text{Fe}_5\text{O}_{12}$ wherein R comprises yttrium and optionally one or more non-fast-relaxing substituents selected from lanthanum, lutetium, bismuth, and gadolinium, and wherein gallium or aluminium substitutes on iron sites in an amount selected to place the saturation magnetization $4\pi M_s$ at a value satisfying $(2.8 \text{ MHz/Oe}) \cdot 4\pi M_s/3 \leq f$ at the stated operating frequency f .
3. The medium of claim 1, being a cryogenic grade characterized at 77 K with $\Delta H \leq 1$ Oe at the stated operating frequency, having fast-relaxing rare-earth content below 10 ppm and the resistivity of claim 1 as a purity proxy.
4. The medium of claim 1, formed as a sphere of diameter d from 2 mm to 15 mm, wherein d , the saturation magnetization $4\pi M_s$, and the stated operating frequency f satisfy $f \geq (2.8 \text{ MHz/Oe}) \cdot 4\pi M_s/3$ and $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$, ϵ_r being the relative permittivity of the body and c the speed of light.

5. The medium of claim 4, polished to an arithmetic-mean roughness $R_a \leq 0.5 \mu\text{m}$ and maximum pit diameter $\leq 0.5 \mu\text{m}$, or, where $\Delta H \leq 1 \text{ Oe}$, to $R_a \leq 0.1 \mu\text{m}$, and bearing a mark of a common crystallographic axis.
6. The medium of claim 1, wherein the large-signal coherence parameter is a threshold drive field at which the uniform precession mode ceases to grow linearly with drive at the stated operating frequency, or a linewidth measured at a stated operating drive amplitude.
7. A method of making the coherence medium of claim 1, comprising: growing a single-crystal iron-garnet boule whose iron-site substituent content is selected from an intended element diameter and operating frequency according to $f \geq (2.8 \text{ MHz/Oe}) \cdot 4\pi M_s/3$ and $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$; orienting the boule and marking a common crystallographic axis; forming and polishing a body to a stated surface class; annealing in oxygen; measuring the linewidth at the operating frequency and grade temperature, the saturation magnetization and resistivity at 295 K, the surface class, and a large-signal coherence parameter; grading the body against the property box; and recording a medium provenance identifying a qualified medium class.
8. The method of claim 7, further comprising measuring the linewidth at 77 K and grading the body as a cryogenic grade only if $\Delta H \leq 1 \text{ Oe}$ at 77 K and fast-relaxing rare-earth content is below 10 ppm.
9. A structured coherence medium for an angular-momentum drive device, comprising a graded, layered, or composite body assembled from a plurality of iron-garnet constituents each having a net uncompensated internal circulation, the body as a whole having a precession- resonance linewidth $\leq 10 \text{ Oe}$ at a stated operating frequency and a resistivity $\geq 10^{10} \Omega \cdot \text{cm}$ at 295 K, the constituents being (a) liquid-phase-epitaxy garnet films of graded gallium content on gadolinium gallium garnet substrates bonded in a stack, or (b) garnet spheres each satisfying $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$ embedded in a low-loss dielectric matrix of loss tangent $\leq 10^{-3}$ at a pitch at which the inter-sphere dipolar field is below the linewidth.
10. A paramagnetic-resonance coherence medium for an angular-momentum drive device, comprising a dilute-paramagnetic single crystal having a paramagnetic-resonance linewidth $\leq 10 \text{ Oe}$ at a stated operating frequency and a resistivity $\geq 10^{10} \Omega \cdot \text{cm}$, together with a stated polarization condition comprising a temperature and frequency at which the thermal polarization $\tanh(h\nu/2kT)$, or a pumped polarization, multiplied by the paramagnetic-center density, yields an aligned moment density of at least a stated floor recorded in the medium provenance.
11. The medium of claim 10, wherein the single crystal is chromium-doped aluminium oxide.
12. A phase-coherent cavity-array drive device comprising: a source frequency-locked by a master absorption lock to a precession resonance of a reference coherence medium; a phase-reference distribution carrying the source phase to N elements; an N-way distribution network with per-branch phase and amplitude adjusters; N resonant cavities each containing a coherence medium having a net uncompensated internal circulation and a linewidth $\leq 10 \text{ Oe}$ at the source frequency;

at each element, an alignment-field source with an alignment-field trim actuator and a coupling probe with a phase detector producing a per-element precession-phase observable relative to the distributed phase reference; and an array controller configured, for each element, to actuate the alignment-field trim actuator so as to hold the element's absorption extremum at the source frequency and to actuate the per-branch phase adjuster so that the element's precession-phase observable lies within a stated tolerance of a commanded phase pattern.

13. The device of claim 12, wherein the alignment-field trim actuator is a trim winding on a per-element bias magnet having a resolution of at most 0.005 Oe, or a thermal trim of a permanent-magnet element, and the array controller actuates it through a per-element dither lock.
14. The device of claim 12, wherein inter-cavity isolation is at least 20 dB, or wherein the cavities form a deliberately coupled chain operated at a band-edge mode of the chain with the per-element alignment-field trim actuators retained.
15. The device of claim 12, wherein the media are spheres each satisfying $f \geq (2.8 \text{ MHz/Oe}) \cdot 4\pi Ms/3$ and $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$ at the source frequency f , mounted with a common marked crystallographic axis at a pitch at which the inter-element dipolar field is below the linewidth.
16. The device of claim 12, wherein the elements are carried on a conformal frame with a common alignment-field bus comprising coil sets or a permanent-magnet lattice with the per-element trim actuators, and the stated tolerance is $\pm 2^\circ$.
17. The device of claim 12, further comprising a memory storing an array allocation map established under sequential, in-phase, phase-scrambled, and inactive-element qualification states, the array controller reporting any assembly-level force offset only from the array allocation map and reporting a null where the map holds no validated nonzero point.
18. A method of operating a phase-coherent cavity-array drive device having N elements each comprising a cavity, a coherence medium, an alignment-field trim actuator, and a coupling probe with a phase detector, the method comprising: energizing an alignment field at a mapped magnitude and sense; acquiring a master absorption lock of a common source on a reference element; for each element, actuating the alignment-field trim actuator to locate the element's absorption extremum at the source frequency and engaging a per-element dither lock; reading each element's precession-phase observable relative to a distributed phase reference and trimming a per-branch phase adjuster until every element lies within a stated tolerance of a commanded phase pattern; tracking resonance drift with the dither locks and reporting out-of-envelope when a trim actuator approaches its range limit; applying a commanded drive envelope within a duty bound while logging a drive-state vector; and reading any assembly-level output only from a stored array allocation map.
19. The method of claim 18, further comprising, on loss of lock at an element, setting that element inactive, re-acquiring its dither lock, and re-verifying its precession-phase observable before returning it to the commanded pattern.

20. A travel-guide feed for a coherence medium in externally driven vertical travel, comprising: a slotted metallic traveling-wave guide having a longitudinal plurality of slot or helix couplers and a matched termination; a dielectric travel tube adjacent the guide within which the medium travels; an alignment-field source spanning the travel with a uniformity of at most one medium linewidth over the travel length, or a plurality of tracking alignment-field trim elements distributed along the travel; a position sensor; and a controller selecting the coupler nearest the medium from the position sensor and holding the medium's precession resonance locked to the source frequency during the travel.
21. The feed of claim 20, wherein the controller commands the tracking trim elements from the position sensor and reads the medium's precession-phase observable through the selected coupler.
22. A method of feeding a coherence medium during externally driven vertical travel, comprising: launching a traveling wave at a source frequency along a slotted guide terminated in a matched load adjacent a dielectric travel tube; maintaining an alignment field along the travel within one linewidth of the resonance field or trimming it from a position sensor as the medium moves; selecting the coupler nearest the medium; and holding the medium's precession resonance locked to the source frequency throughout the travel.
23. A mounting assembly for a coherence medium comprising a low-loss dielectric post, a strain-decoupling throat between the post and the medium, a minimal-area bond to the medium, and a support interface transmitting the medium's support force to a force sensor or load-bearing structure, the throat being dimensioned such that the mechanical stress transmitted to the medium over the operating temperature range produces a magnetoelastic shift of the medium's precession resonance, approximately $3\lambda\sigma/Ms$, smaller than the medium's linewidth, and such that magnetostrictive deformation of the medium transmits substantially no steady force to the support interface.
24. The mounting assembly of claim 23, wherein the post is sapphire or alumina and forms a thermal path from the medium to a baseplate having a conductance of at least 0.02 W/K.
25. A method of mounting a coherence medium, comprising: bonding the medium through a minimal-area bond to a strain-decoupling throat of a low-loss dielectric post; measuring the medium's precession-resonance field before and after bonding and over the operating temperature range; and accepting the mount only if the shift is smaller than the medium's linewidth.
26. A sealed drive module comprising: a drive-transparent structural housing; a coherence medium having a net uncompensated internal circulation, a linewidth ≤ 10 Oe at a stated operating frequency, and a resistivity $\geq 10^{10}$ $\Omega\cdot\text{cm}$; a resonant cavity; a mounting assembly carrying the medium and comprising a low-loss dielectric post with a strain-decoupling throat and a minimal-area bond; a bias magnet with trim windings; a drive chain comprising a modulator with a controller-enforced duty bound and incident, reflected, and transmitted power detectors; a controller and sense unit behind a shield partition; non-magnetic fasteners; a first thermal path from the drive chain and windings to a baseplate; a second thermal path from the medium through

the post of the mounting assembly to the baseplate, having a stated thermal conductance; a calibration memory holding a stored empirical response map comprising a medium provenance, a lock curve, a force-offset transfer map that admits zero, a thermal drift map, a validity envelope with runtime limits, and safe-off parameters; runtime witnesses comprising at least medium temperature, alignment field, lock error, probe phase, absorbed power and duty, and support-axis field gradient; and an interface set comprising a mounting datum, DC power, at least one RF or distributed-feed port, a data and control bus reporting a drive-state vector, a validity state, and a map identifier, and a thermal interface at the baseplate.

27. The module of claim 26, wherein the second thermal path is a sapphire or alumina post of conductance at least 0.02 W/K, and the controller enforces a medium duty envelope such that the product of the medium temperature rise and the resonance temperature coefficient remains below the medium linewidth.
28. The module of claim 26, wherein the controller maintains validity states comprising valid, degraded, out-of-envelope, invalid, and safe-off, inhibits any force-setpoint mode unless the stored map holds at least one validated nonzero operating point, and otherwise operates the module as a null reference drive device.
29. The module of claim 26, wherein the housing is rated for vacuum or dry-gas fill and fragment retention, and the interface set further comprises a support-path force-sensor port.
30. A method of operating the sealed drive module of claim 26, comprising: loading the stored empirical response map and verifying that its provenance matches the module identifier; performing a drive-off zero check; entering a valid state only if every runtime witness lies within the runtime limits; locking the drive chain to the medium's precession resonance; applying a commanded drive envelope within the duty bound; reporting any force offset only from the force-offset transfer map with its uncertainty; interleaving drive-off zero checks; and transitioning to safe-off on loss of lock, a runtime-limit exceedance, or a map check failure.
31. A method of factory-qualifying a drive module comprising a coherence medium, an alignment-field source, a cavity, and a drive chain, the method comprising: supporting the module through its support interface on a force transducer having a stated noise floor R over a stated averaging interval, while carrying the alignment-field source and feeds on a reaction frame having a separate load path; locking the drive to the medium's precession resonance and recording a lock curve; executing, in randomized settled blocks, drive-on and drive-off states, a heating-matched detuned state in which module dissipation and medium temperature history are matched to the resonant state, a full-system field reversal in which the bias source and every trim element are reversed, a bias-only reversal, a moment-modulation dummy state in which a magnetically equivalent body of equal static moment is driven to reproduce the resonant change of moment, and a gradient scan in which deliberate support-axis gradients of both signs are applied; computing differential channels and their noise floors; applying validity gates comprising repeatability, resonance contrast against the heating-matched state, moment-equivalent material contrast, directional signature by full-

system reversal compared with bias-only reversal, gradient exclusion by the intercept of the gradient scan, thermal exclusion by time-class separation, mechanical and servo exclusion, and statistical validity at $\geq 3R$; recording a force-offset transfer map that admits zero, a channel sign at each validated point, and a thermal drift map; and writing to the module's calibration memory a ship record comprising medium provenance, lock curve, transfer map, drift map, noise floor, gate results, validity envelope, safe-off parameters, and a status of qualified module or null reference.

32. The method of claim 31, wherein the thermal exclusion comprises modulating the drive at a frequency above the measured thermal corner of the medium and mount and synchronously demodulating the force channel, and the mechanical and servo exclusion comprises repeating the measurement with the medium at two distances from the force transducer mechanism.
33. The method of claim 31, wherein the gradient exclusion further comprises mounting two media with antiparallel alignment in a common gradient and comparing their combined channel with the sum of their individual channels.
34. The method of claim 31, further comprising qualifying a production class by reproducing the registered channel sign and transfer map on a second jig, unit, or run series within uncertainty, and accepting further units under the class map only after an acceptance test at one or more validated points against a separate force reference.
35. The method of claim 31, applied to an assembled array of modules, further comprising executing sequential, in-phase, phase-scrambled, and inactive-element states with per-element and assembly-level force channels and recording an array allocation map with cross-talk terms, without presuming that assembly-level force offset is the sum of per-element offsets.
36. A tiled coherence-media assembly comprising a plurality of coherence media each having a net uncompensated internal circulation, a linewidth ≤ 10 Oe at a common source frequency, a resistivity $\geq 10^{10} \Omega \cdot \text{cm}$, and, where spherical, satisfying $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$ at that frequency, mechanically integrated in a common matrix or frame with individual coupling structures and a common alignment-field bus having per-element trim elements, the assembly presenting an aggregate number of net aligned moments exceeding that of any constituent medium and being driven from a common locked source through an N-way distribution network with per-element phase adjusters and per-element precession-phase observables.
37. The assembly of claim 36, wherein the aggregate driven-medium mass is at least 0.1 kg; at least 1 kg; at least 100 kg; or at least 1000 kg, the media being tiled elements each within the design rule and none being a single continuous body exceeding the design rule.
38. The assembly of claim 36, further comprising a memory storing an array allocation map established under sequential, in-phase, phase-scrambled, and inactive-element states, and a controller reporting any assembly-level force offset only from that map.

39. A method of operating the tiled assembly of claim 36, comprising: acquiring a master absorption lock of the common locked source on a reference element; for each element, actuating its trim element to hold its absorption extremum at the source frequency under a per-element dither lock; trimming each per-element phase adjuster until the element's precession-phase observable lies within a stated tolerance of a commanded phase pattern; applying a commanded drive envelope within a duty bound; and reading any assembly-level output only from a stored array allocation map established under sequential, in-phase, phase-scrambled, and inactive-element qualification states.
40. A prophetic POAMS-native coherence medium for an angular-momentum device, comprising a structured, extended, topological, or collective medium having a measurable internal circulation, a drive-addressable coherent phase, and a whole-turn-supporting state, wherein the medium is not limited to the magnetostatic uniform-mode size-frequency condition of a YIG sphere and is qualified under a property box appropriate to its own mode, geometry, temperature, and readout.

ABSTRACT

Engineered coherence media, phase-coherent cavity-array drive devices, travel-guide feeds, stress-isolating mounts, sealed drive modules, and tiled media assemblies for angular-momentum drive devices, with methods of making, qualifying, and operating them. Media are single-crystal iron-garnet bodies specified by a property box under stated measurement conditions—linewidth, resistivity, saturation, surface class, large-signal coherence—and a size–frequency–saturation design rule, $f \geq (2.8 \text{ MHz/Oe}) \cdot 4\pi \text{ Ms}/3$ and $d \cdot f \cdot \sqrt{\epsilon_r} \leq c/10$, with cryogenic grades, structured species, and a separately claimed prophetic paramagnetic-resonance class. Arrays distribute one locked source across N cavities with per-element alignment-field trim and feed-phase loops closed on a per-element precession-phase observable. Modules carry split thermal paths, a calibration memory with a force-offset transfer map that admits zero, validity states, and safe-off. Factory qualification applies gates E1–E9 with heating-matched detuning, moment-equivalent dummies, full-system reversal, gradient scans, and time-class separation, shipping every module as a qualified module or null reference.

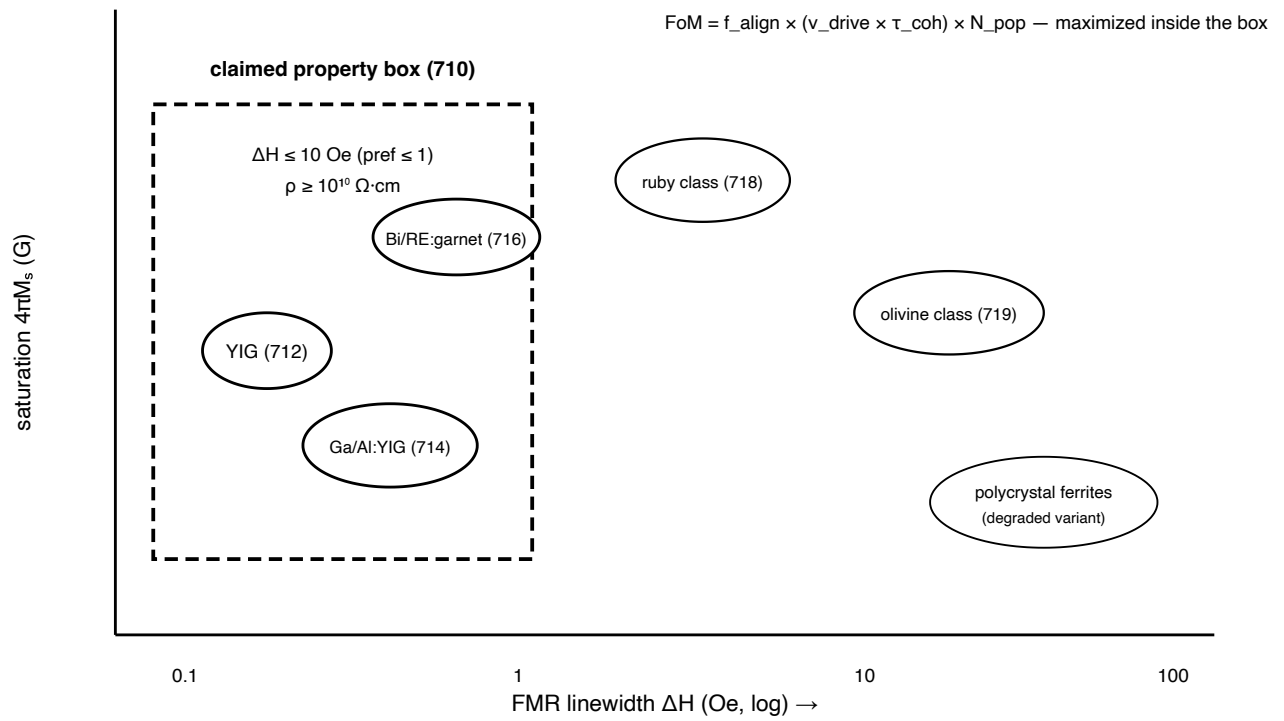
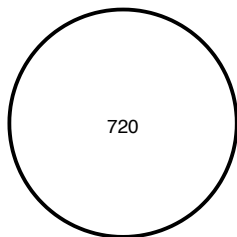


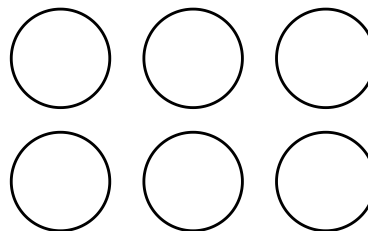
FIG. 1

Figure-of-merit map across composition space; the claimed media property box.

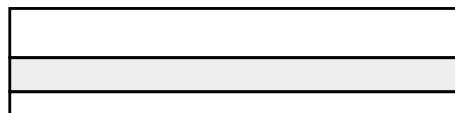


single sphere $\varnothing$ 2–15 mm
surface $\leq 0.5 \mu\text{m}$ (722)

uniform internal-field response (extant: demagnetizing factor $N = \frac{1}{2}$)
→ single sharp resonance



sphere array (724) — plural controlled elements



layered / graded media (726)
spatially shaped drive response

FIG. 2

Medium geometries — sphere, array, graded/layered forms.

SIZE-FREQUENCY-SATURATION DESIGN RULE (700)

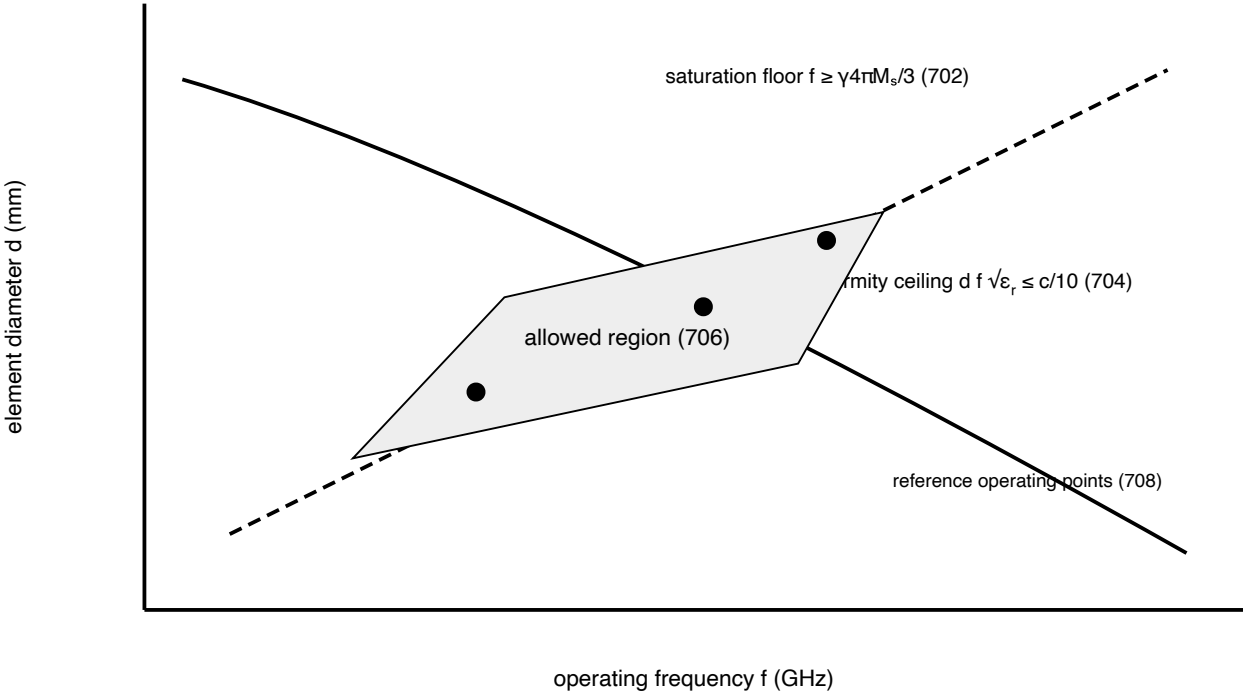
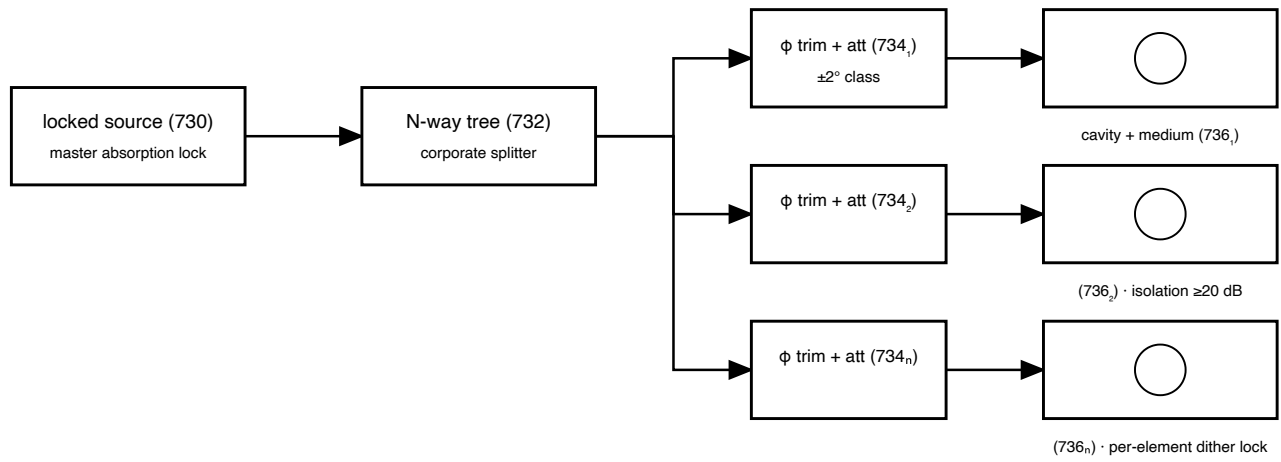


FIG. 3

Size-frequency-saturation design-rule chart with saturation floor, uniformity ceiling, allowed region, and reference operating points.



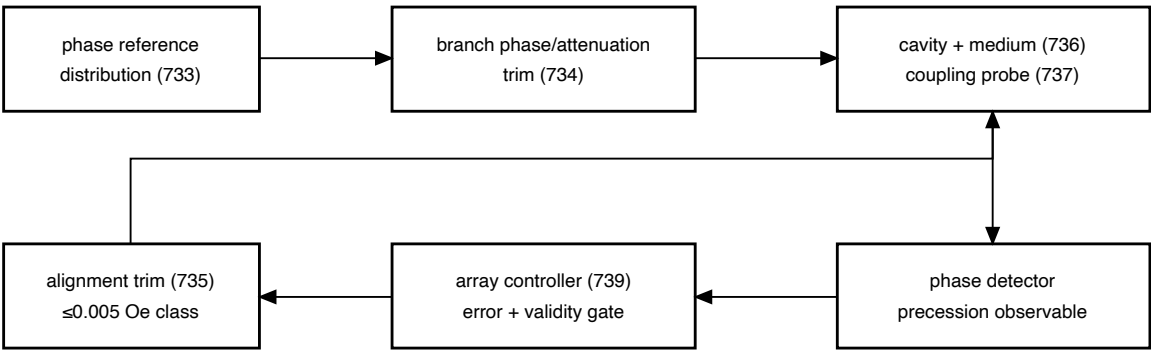
: assembly output read from measured array allocation map; additivity not presumed :

alternative: deliberately coupled cavity chain operated as one extended mode (738)

FIG. 4

Phase-coherent N-element cavity-array drive device with per-element phase, bias, probe, and lock control.

PER-ELEMENT CONTROL AND LOCK ACQUISITION



acquire resonance → measure probe phase → trim bias and feed → validate isolation → store allocation map

runtime: maintain phase within the qualified envelope; inhibit an element on loss of lock
additivity is measured, not presumed

FIG. 5

Per-element control loop and array lock-acquisition sequence.

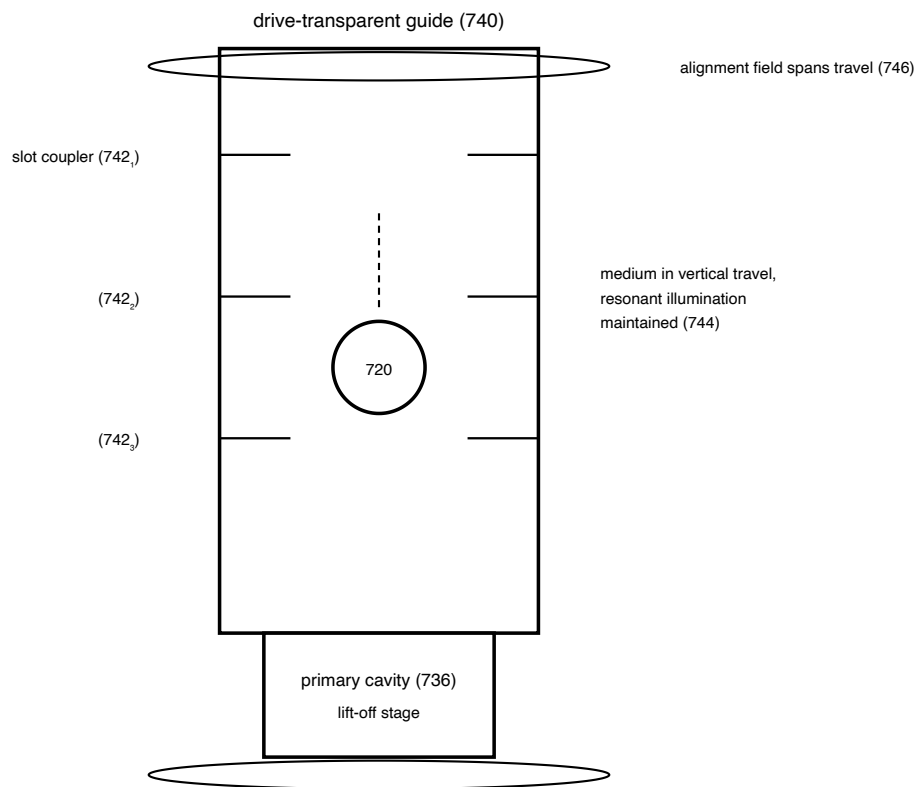


FIG. 6

Travel-guide feed — distributed couplers, tracking trim, and position sensing along externally driven travel.

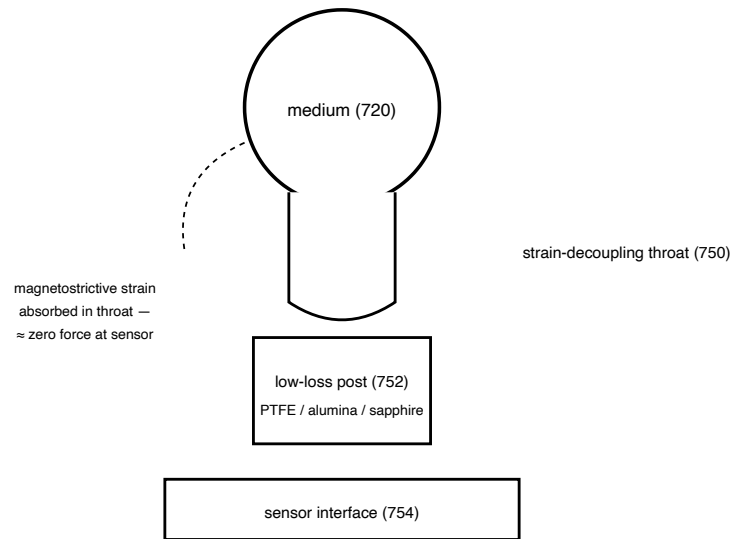


FIG. 7

Stress-isolating, artifact-nulled mount with magnetoelastic budget and high-conductivity post option.

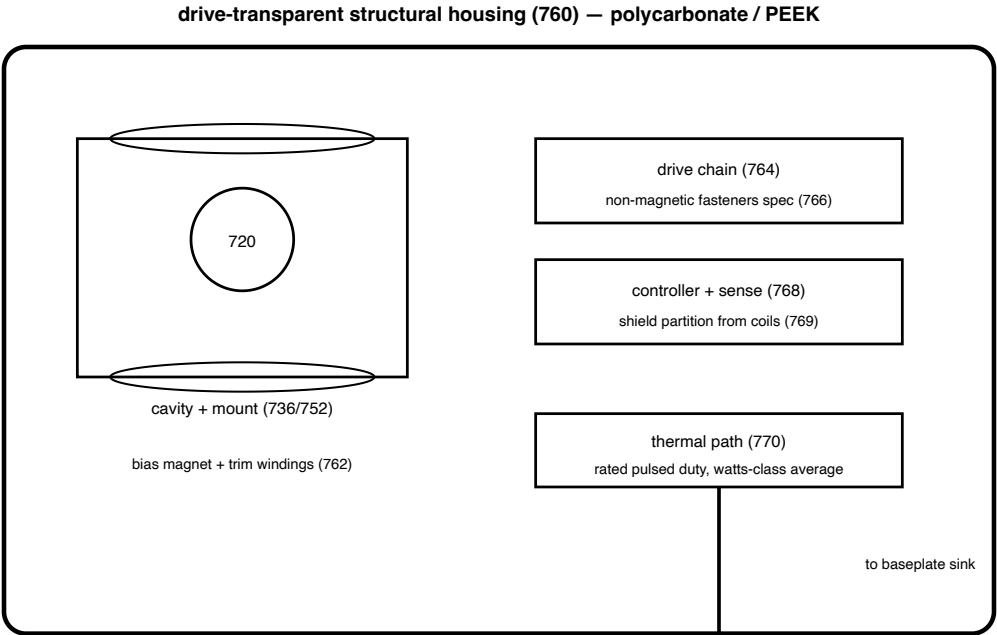
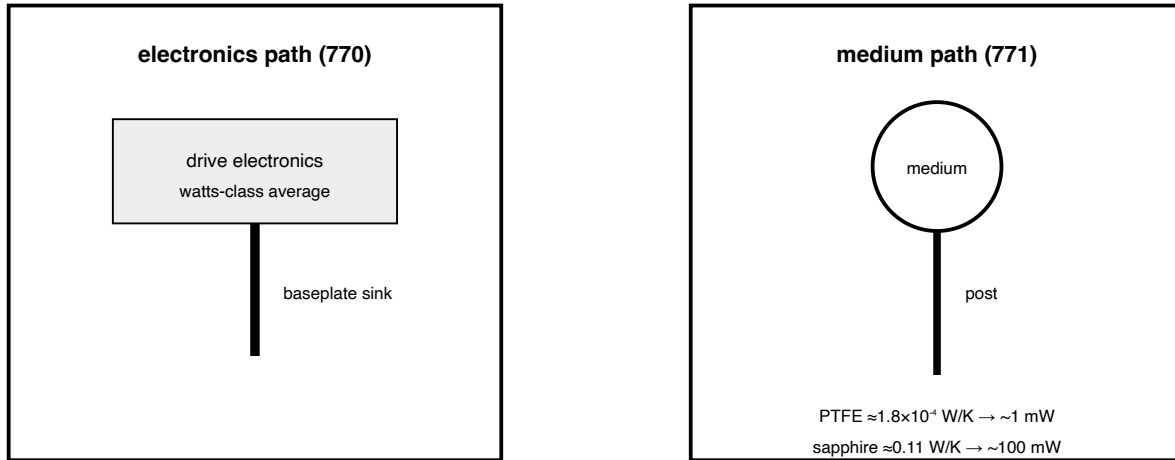


FIG. 8

Sealed drive module with separate electronics and medium thermal paths, calibration memory, witnesses, interfaces, and safe-off.

SPLIT THERMAL PATHS



separate budgets prevent electronics heat from being assigned to the medium support

FIG. 9

Separate electronics and medium thermal paths with representative conductance arithmetic.

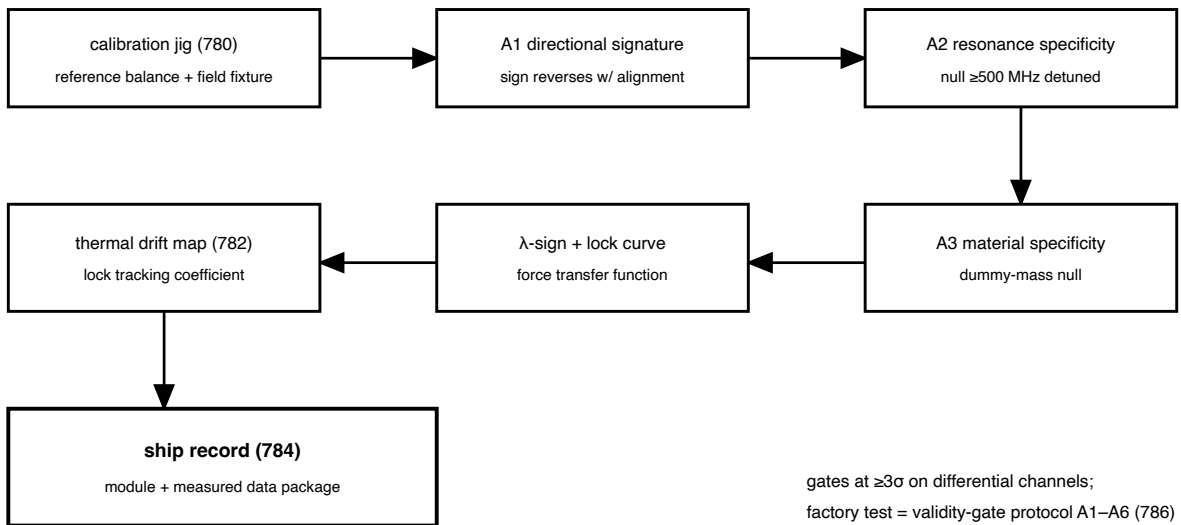


FIG. 10
Factory qualification jig and gates E1–E9 producing a provenance-bearing ship record.

DISCRIMINATING CONTROLS

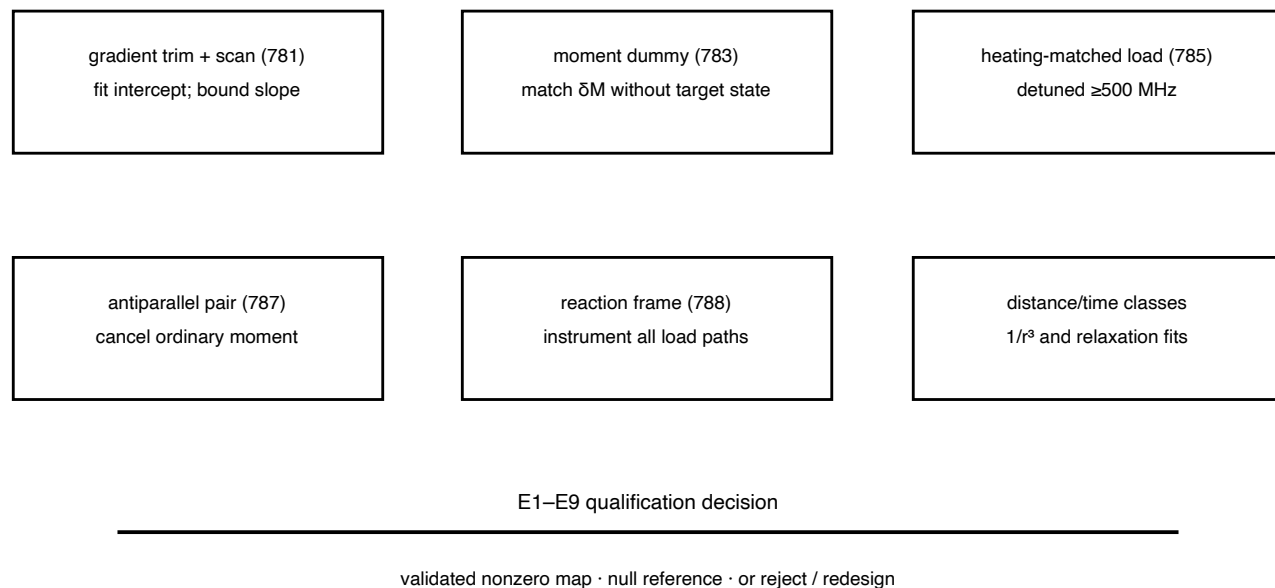
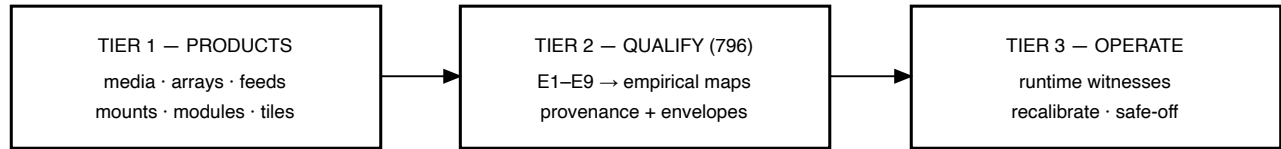


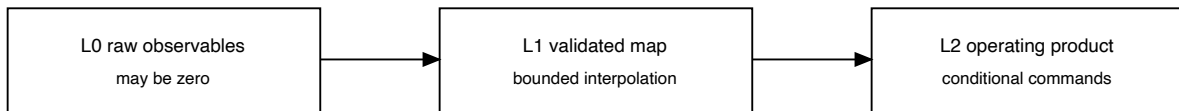
FIG. 11

Gradient, moment, heating, antiparallel, reaction-path, distance, and time-class controls.

THREE TIERS · LAYERED SCOPE · VALIDITY STATES



LAYERED SCOPE (794)



VALIDITY STATE MACHINE (798)

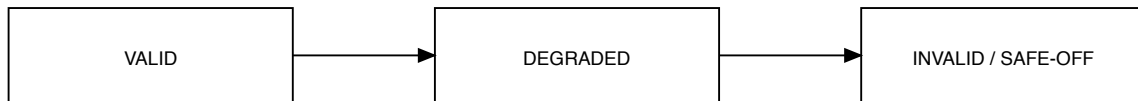


FIG. 12

Three-tier architecture, layered scope L0–L2, and validity-state machine.